



Kristoffer Nordström

Embedded Systems, FPGA & HIL Verification Engineer

 April 1st, 1985
 nordstroem.ch
 kristoffer@nordstroem.ch

Languages

 German ● ● ● ● ●
 English ● ● ● ● ●
 Swedish ● ● ● ● ●

Technology

 Python, pytest, uv
 VHDL, VUnit, PSL, P-org
 C/C++, Bash, MATLAB
 Astro/TS, PHP, SQL

Tooling

 Codebeamer ALM
 Jenkins, GitHub Actions
 git, Docker, Ansible

Networks

 kown7
 kristoffer-nordström-a1565277

Introduction

Senior embedded systems, FPGA and verification engineer with more than 15 years of experience building and validating safety-critical systems at the intersection of hardware, firmware and software.

Strong background in requirements-driven development, FPGA-based test harnesses, hardware-in-the-loop (HIL) verification, Linux drivers, Python/pytest automation, CI integration and verification.

I am particularly interested in applying this experience to create robust subsystem validation and reliable integration testing to enable complex real-world systems. My strongest contribution is building the verification infrastructure around embedded controllers, FPGA-based interfaces, sensor and actuator loops, safety mechanisms and automated regression testing before full-system integration. I thrive on finding simple solutions outside the box.

Work Experience

Present **Development Engineer** Super Computing Systems AG, Zürich
Test engineer for subsystems of a railway interlocking system. These interconnected subsystems communicate over Ethernet or direct traces and contain FPGA control modules. Their firmware was developed according to the EN 50716 standard.

Key Contributions

- Derived test specifications from subsystem requirements and safety classifications.
- Designed and implemented FPGA-based hardware-in-the-loop test harnesses using PCIe on evaluation-kits with custom VHDL and Linux drivers.
- Implemented automated test cases/framework in Python and orchestrated them with pytest.
- Performed verification tasks for the document driven development process.

Technology Ambassador

Advisory role supporting management with strategic technology decisions, engineering tooling and internal technology development.

Key Contributions

- supporting the technology roadmap, e.g., high-level synthesis and test automation with VUnit
- migrating away from legacy Atlassian tooling into the cloud
- optimizing license usage in FPGA development
- organizing the annual internal *Hackday*

2017

2017 **Development Engineer** nanoTronic GmbH, Lyss
Software engineer in a customer project. I upgraded an Ethernet and STM32-based legacy application to a wireless solution on an ESP32 chip. I was mainly responsible for refactoring the C/C++ codebase and aligning requirements with the customer.

2015

FPGA firmware verification engineer for an ESM/Orion subsystem for Artemis I. I was responsible for ensuring that the firmware conformed to customer specifications (ECSS) and to requirements derived in-house.

Keywords

- Embedded systems
- FPGA
- VHDL
- HIL
- hardware-in-the-loop
- verification
- validation
- safety-critical systems
- SIL 4
- EN 50716
- ECSS
- robotics verification
- test automation
- CI
- pytest
- Linux drivers
- PCIe
- Ethernet
- GPIO
- requirements engineering
- traceability
- regression testing

- 2014
- FPGA Development Engineer
- Super Computing Systems AG, Zürich
- For an application in biochemical analysis:
- Development and verification of data and control paths for a high-speed digital receiver in VHDL
- 2014
- Porting existing code to a Xilinx 7 Series FPGA

Education

- 2013
- MSc Electrical Engineering & Information Technology
- ETH Zürich
- Focus: *Microelectronics and Communication*
- Master's thesis: *Tuneable Operational Transconductance Amplifier in Deep Submicron CMOS*. A tunable OTA for an LTE+ baseband filter was successfully designed in a 28 nm node.
- Semester thesis: *Turbo Decoder Implementation for Evolved EDGE*. Group project for a standalone turbo decoder ASIC for Evolved EDGE. The chip was manufactured with a UMC 0.18 µm node and successfully tested. See C. Benkeser et al., *Turbo decoder design for high code rates*, 2012.)
- 2011
- BSc Electrical Engineering & Information Technology
- ETH Zürich
- Major: *Micro- and Optoelectronics*
- 2011
- Erasmus exchange at Lund's LTH, 2008/2009. Part of [FSAE LUR3](#).
- 2006

Other Certifications

- 2024
- Black Box Software Testing Foundations
- BBST
- 2023
- The Detailed Design Clinic
- Michael Monty Montgomery, iDesign
- 2020
- Project Design Master Class
- Juval Löwy, iDesign
- 2018
- Architect's Master Class
- Juval Löwy, iDesign

Other Work

- Present
- Maintainer
- sltoo.dev
- I added traceability features to *rmtoo*. I presented these features as an accepted speaker at SWISSED21 and FOSDEM'22 ([video](#))
- 2018
- 2024
- Developer
- Exercise Generator
- A proof-of-concept to automate the generation of math exercises for German universities.
- 2023

References

Available upon request.